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Forage Quantium Data Analytics Task 1 : Data Preparation and Customer Analytics

1. Importing the necessary dependencies

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import plotly as plt
import plotly.graph_objects as go
import plotly.express as px
import seaborn as sns
from IPython.display import IFrame, display
```

2. Reading the datasets using the pandas module

In [2]:
```
pur_beh = pd.read_csv('QVI_purchase_behaviour.csv')
tran = pd.read_excel('QVI_transaction_data.xlsx')
```

In [3]:
```
pur_beh.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 72637 entries, 0 to 72636
Data columns (total 3 columns):
 #   Column            Non-Null Count  Dtype 
---  ------            --------------  ----- 
 0   LYLTY_CARD_NBR    72637 non-null  int64 
 1   LIFESTAGE         72637 non-null  object
 2   PREMIUM_CUSTOMER  72637 non-null  object
dtypes: int64(1), object(2)
memory usage: 1.7+ MB
```

In [4]:
```
tran.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264836 entries, 0 to 264835
Data columns (total 8 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
 0   DATE            264836 non-null  int64
 1   STORE_NBR       264836 non-null  int64
 2   LYLTY_CARD_NBR  264836 non-null  int64
 3   TXN_ID          264836 non-null  int64
 4   PROD_NBR        264836 non-null  int64
 5   PROD_NAME       264836 non-null  object
 6   PROD_QTY        264836 non-null  int64
 7   TOT_SALES       264836 non-null  float64
dtypes: float64(1), int64(6), object(1)
memory usage: 16.2+ MB
```

From above we can infer that there are no Blank/NULL VALUES in both the datasets

3. Exploring the Purchase Behavior Dataset

In [5]:
```
pur_beh.LIFESTAGE.value_counts()
```

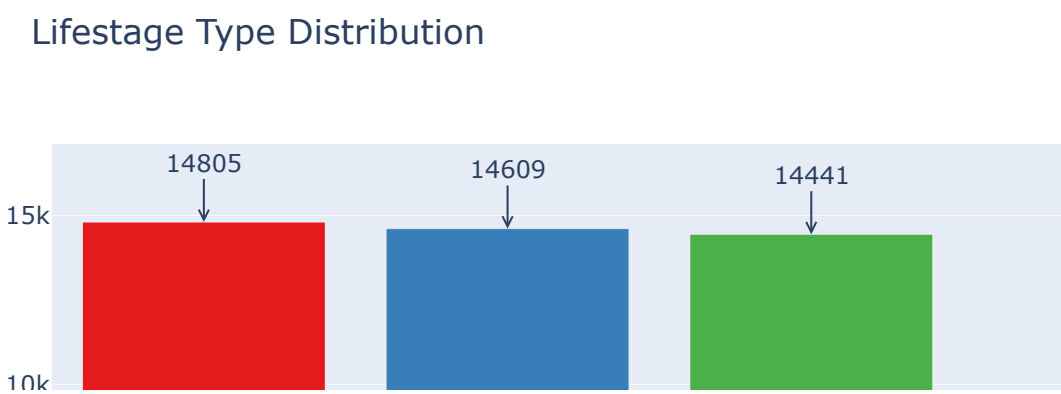
Out[5]:
```
RETIREES                  14805
OLDER SINGLES/COUPLES     14609
YOUNG SINGLES/COUPLES     14441
OLDER FAMILIES             9780
YOUNG FAMILIES             9178
MIDAGE SINGLES/COUPLES     7275
NEW FAMILIES               2549
Name: LIFESTAGE, dtype: int64
```

In [6]:
```
pur_beh.PREMIUM_CUSTOMER.value_counts()
```

Out[6]:
```
Mainstream    29245
Budget        24470
Premium       18922
Name: PREMIUM_CUSTOMER, dtype: int64
```

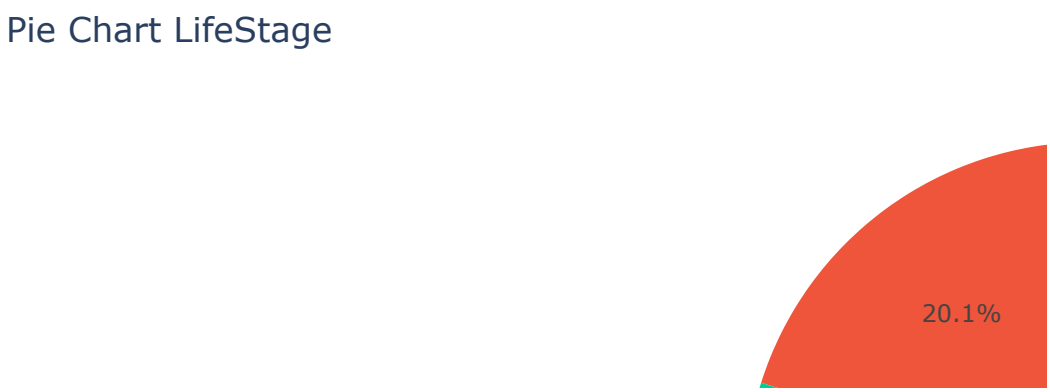
In [7]:
```
lifestage_type_counts = pur_beh['LIFESTAGE'].value_counts()

# Define custom colors for each lifestage type
colors = px.colors.qualitative.Set1[:len(lifestage_type_counts)]

# Create a bar chart using Plotly Express with different colors
fig = px.bar(
    x=lifestage_type_counts.index,
    y=lifestage_type_counts.values,
    labels={'y': 'Count', 'x': 'LifeStage'},
    color=lifestage_type_counts.index,  # Use lifestage type as color
    color_discrete_map={ctype: color for ctype, color in zip(lifestage_type_counts.
    title='Lifestage Type Distribution',
)
for i, count in enumerate(lifestage_type_counts.values):
```

```
    fig.add_annotation(
        x=lifestage_type_counts.index[i],
        y=count,
        text=str(count),
        showarrow=True,
        arrowhead=5,
        ax=0,
        ay=-30,
    )

# Add a legend
fig.update_layout(legend=dict(title=dict(text='Legend')))

# Show the plot
fig.show()
```



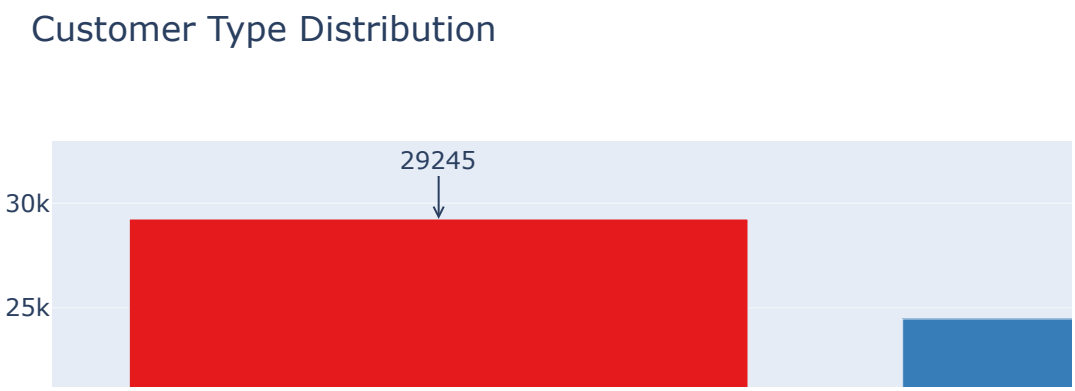
In [8]:
```
labels = pur_beh['LIFESTAGE'].value_counts().index
values = pur_beh['LIFESTAGE'].value_counts().values
fig = go.Figure(data=[go.Pie(labels=labels, values=values)])

# Set layout properties
fig.update_layout(title='Pie Chart LifeStage')
```

```
# Show the plot
fig.show()
```



In [9]:
```
premium_type_counts = pur_beh['PREMIUM_CUSTOMER'].value_counts()

# Define custom colors for each premium type
colors = px.colors.qualitative.Set1[:len(premium_type_counts)]

# Create a bar chart using Plotly Express with different colors
fig = px.bar(
    x=premium_type_counts.index,
    y=premium_type_counts.values,
    labels={'y': 'Count', 'x': 'Type of Customer'},
    color=premium_type_counts.index,  # Use premium type as color
    color_discrete_map={ctype: color for ctype, color in zip(premium_type_counts.in
    title='Customer Type Distribution',
)
for i, count in enumerate(premium_type_counts.values):
    fig.add_annotation(
        x=premium_type_counts.index[i],
        y=count,
        text=str(count),
        showarrow=True,
        arrowhead=5,
```

```
        ax=0,
        ay=-30,
    )

# Add a legend
fig.update_layout(legend=dict(title=dict(text='Legend')))

# Show the plot
fig.show()
```



In [10]:
```
labels = pur_beh['PREMIUM_CUSTOMER'].value_counts().index
values = pur_beh['PREMIUM_CUSTOMER'].value_counts().values
fig = go.Figure(data=[go.Pie(labels=labels, values=values)])

# Set layout properties
fig.update_layout(title='Pie Chart Customer Type')

# Show the plot
fig.show()
```

Pie Chart Customer Type

4. Exploring the Transactions Dataset

In [11]:
```
tran.describe()
```

Out[11]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PR
count	264836.000000	264836.00000	2.648360e+05	2.648360e+05	264836.000000	26483
mean	43464.036260	135.08011	1.355495e+05	1.351583e+05	56.583157	
std	105.389282	76.78418	8.057998e+04	7.813303e+04	32.826638	
min	43282.000000	1.00000	1.000000e+03	1.000000e+00	1.000000	
25%	43373.000000	70.00000	7.002100e+04	6.760150e+04	28.000000	
50%	43464.000000	130.00000	1.303575e+05	1.351375e+05	56.000000	
75%	43555.000000	203.00000	2.030942e+05	2.027012e+05	85.000000	
max	43646.000000	272.00000	2.373711e+06	2.415841e+06	114.000000	20

In [12]:
```
tran
```

Out[12]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD
0	43390	1	1000	1	5	Natural Chip Compny SeaSalt175g	
1	43599	1	1307	348	66	CCs Nacho Cheese 175g	
2	43605	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
3	43329	2	2373	974	69	Smiths Chip Thinly S/Cream&Onion 175g	
4	43330	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	
...	...	...	...	...	...	...	
264831	43533	272	272319	270088	89	Kettle Sweet Chilli And Sour Cream 175g	
264832	43325	272	272358	270154	74	Tostitos Splash Of Lime 175g	
264833	43410	272	272379	270187	51	Doritos Mexicana 170g	
264834	43461	272	272379	270188	42	Doritos Corn Chip Mexican Jalapeno 150g	
264835	43365	272	272380	270189	74	Tostitos Splash Of Lime 175g	

264836 rows × 8 columns

5. Combining both the datasets on basis of common column named 'LYLTY_CARD_NBR'

In [13]:
```
merged_data = pd.merge(pur_beh, tran, on='LYLTY_CARD_NBR', how='inner')
```

In [14]:
```
merged_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 264836 entries, 0 to 264835
Data columns (total 10 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   LYLTY_CARD_NBR    264836 non-null  int64
 1   LIFESTAGE         264836 non-null  object
 2   PREMIUM_CUSTOMER  264836 non-null  object
 3   DATE              264836 non-null  int64
 4   STORE_NBR         264836 non-null  int64
 5   TXN_ID            264836 non-null  int64
 6   PROD_NBR          264836 non-null  int64
 7   PROD_NAME         264836 non-null  object
 8   PROD_QTY          264836 non-null  int64
 9   TOT_SALES         264836 non-null  float64
dtypes: float64(1), int64(6), object(3)
memory usage: 22.2+ MB
```

In [15]:
```
merged_data.LIFESTAGE.value_counts()
```

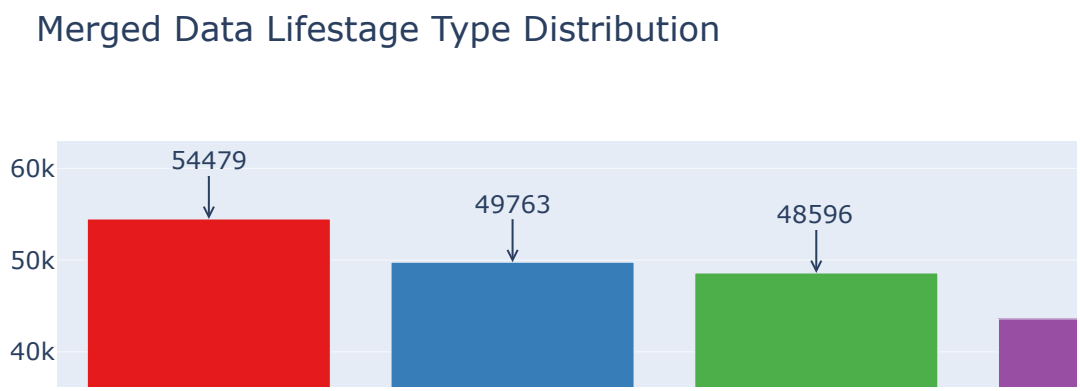
Out[15]:
```
OLDER SINGLES/COUPLES     54479
RETIREES                  49763
OLDER FAMILIES            48596
YOUNG FAMILIES            43592
YOUNG SINGLES/COUPLES     36377
MIDAGE SINGLES/COUPLES    25110
NEW FAMILIES               6919
Name: LIFESTAGE, dtype: int64
```

In [16]:
```
merged_data.PREMIUM_CUSTOMER.value_counts()
```

Out[16]:
```
Mainstream    101988
Budget         93157
Premium        69691
Name: PREMIUM_CUSTOMER, dtype: int64
```

In [17]:
```
lifestage_type_counts = merged_data['LIFESTAGE'].value_counts()

# Define custom colors for each lifestage type
colors = px.colors.qualitative.Set1[:len(lifestage_type_counts)]

# Create a bar chart using Plotly Express with different colors
fig = px.bar(
    x=lifestage_type_counts.index,
    y=lifestage_type_counts.values,
    labels={'y': 'Count', 'x': 'LifeStage'},
    color=lifestage_type_counts.index,  # Use lifestage type as color
    color_discrete_map={ctype: color for ctype, color in zip(lifestage_type_counts.
    title='Merged Data Lifestage Type Distribution',
)
for i, count in enumerate(lifestage_type_counts.values):
    fig.add_annotation(
        x=lifestage_type_counts.index[i],
        y=count,
        text=str(count),
        showarrow=True,
```

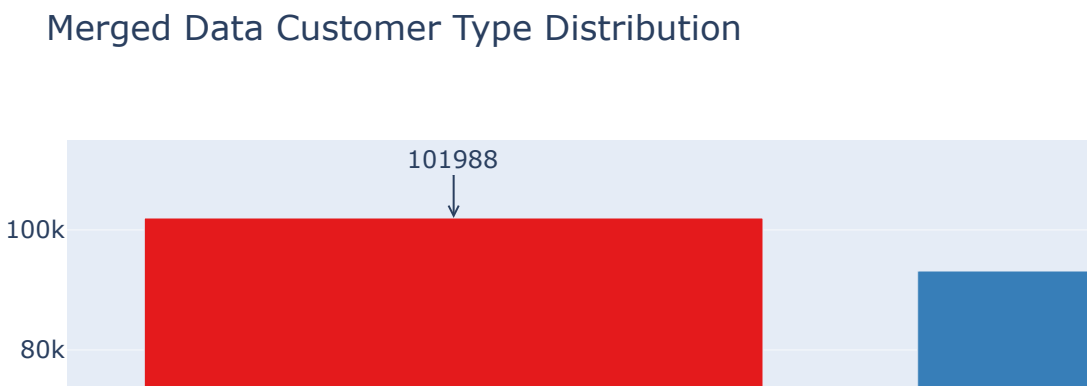
```
        arrowhead=5,
        ax=0,
        ay=-30,
    )

# Add a legend
fig.update_layout(legend=dict(title=dict(text='Legend')))

# Show the plot
fig.show()
```



In [18]:
```
labels = merged_data['LIFESTAGE'].value_counts().index
values = merged_data['LIFESTAGE'].value_counts().values
fig = go.Figure(data=[go.Pie(labels=labels, values=values)])

# Set layout properties
fig.update_layout(title='Pie Chart Merged Data LifeStage')

# Show the plot
fig.show()
```

In [19]:
```
premium_type_counts = merged_data['PREMIUM_CUSTOMER'].value_counts()

# Define custom colors for each premium type
colors = px.colors.qualitative.Set1[:len(premium_type_counts)]

# Create a bar chart using Plotly Express with different colors
fig = px.bar(
    x=premium_type_counts.index,
    y=premium_type_counts.values,
    labels={'y': 'Count', 'x': 'Type of Customer'},
    color=premium_type_counts.index,  # Use premium type as color
    color_discrete_map={ctype: color for ctype, color in zip(premium_type_counts.in
    title='Merged Data Customer Type Distribution',
)
for i, count in enumerate(premium_type_counts.values):
    fig.add_annotation(
        x=premium_type_counts.index[i],
        y=count,
        text=str(count),
        showarrow=True,
        arrowhead=5,
        ax=0,
        ay=-30,
    )
```

```
# Add a legend
fig.update_layout(legend=dict(title=dict(text='Legend')))

# Show the plot
fig.show()
```



In [20]:
```
labels = merged_data['PREMIUM_CUSTOMER'].value_counts().index
values = merged_data['PREMIUM_CUSTOMER'].value_counts().values
fig = go.Figure(data=[go.Pie(labels=labels, values=values)])

# Set layout properties
fig.update_layout(title='Pie Chart Merged Data Customer Type')

# Show the plot
fig.show()
```

Pie Chart Merged Data Customer Type

In [21]:
```
#Formating the DATE COLUMN from Microsoft Excel format to Normal Format
merged_data['DATE_NRML'] = pd.to_datetime(merged_data['DATE'], origin='1900-01-01',
```

In [22]:
```
#Extracting Year from DATE_NRML column
merged_data['Year'] = merged_data['DATE_NRML'].dt.year
```

In [23]:
```
#Extracting Month_name from DATE_NRML column
merged_data['Month_Name'] = merged_data['DATE_NRML'].dt.strftime('%B')
```

In [24]:
```
#Extracting Month_name and Year from DATE_NRML column
merged_data['Month_Year'] = merged_data['DATE_NRML'].dt.strftime('%B %Y')
```

In [25]:
```
#Extracting Week Number from DATE_NRML column
merged_data['Week_Number'] = merged_data['DATE_NRML'].dt.isocalendar().week
```

In [26]:
```
##Extracting Quater from DATE_NRML column
merged_data['Quarter'] = merged_data['DATE_NRML'].dt.quarter
```

In [27]:
```
#Extracting Quater and Year from DATE_NRML column
merged_data['Quarter_Year'] = merged_data['DATE_NRML'].dt.to_period('Q')
```

In [28]:
```
#Extracting WEEKDAY from DATE_NRML column
merged_data['Weekday'] = merged_data['DATE_NRML'].dt.day_name()
```

In [29]:
```
merged_data.head(10)
```

Out[29]:

	LYLTY_CARD_NBR	LIFESTAGE	PREMIUM_CUSTOMER	DATE	STORE_NBR	TXN_ID
0	1000	YOUNG SINGLES/COUPLES	Premium	43390	1	1
1	1002	YOUNG SINGLES/COUPLES	Mainstream	43359	1	2
2	1003	YOUNG FAMILIES	Budget	43531	1	3
3	1003	YOUNG FAMILIES	Budget	43532	1	4
4	1004	OLDER SINGLES/COUPLES	Mainstream	43406	1	5
5	1005	MIDAGE SINGLES/COUPLES	Mainstream	43462	1	6
6	1007	YOUNG SINGLES/COUPLES	Budget	43438	1	7
7	1007	YOUNG SINGLES/COUPLES	Budget	43439	1	8
8	1009	NEW FAMILIES	Premium	43424	1	9
9	1010	YOUNG SINGLES/COUPLES	Mainstream	43352	1	10

In [30]:
```
merged_data.to_csv('Merged_data.csv')
#Saving the merged Dataset
```

In [31]:
```
merged_data.head(10)
```

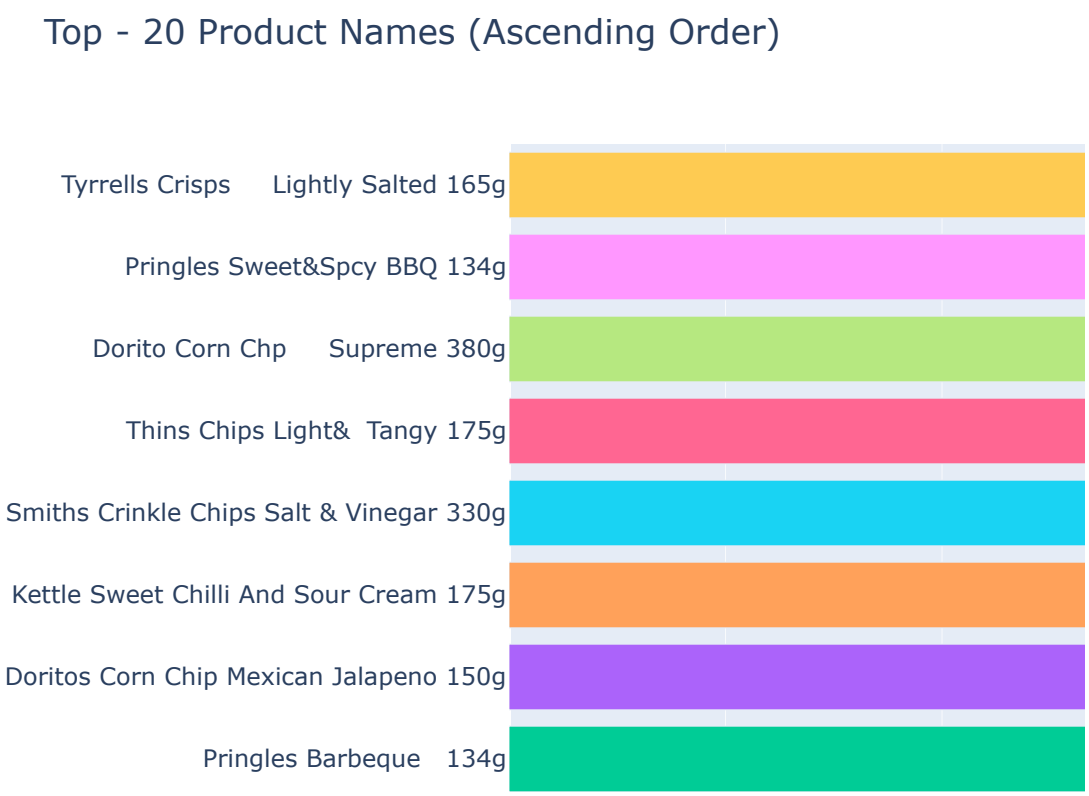
Out[31]:

	LYLTY_CARD_NBR	LIFESTAGE	PREMIUM_CUSTOMER	DATE	STORE_NBR	TXN_ID
0	1000	YOUNG SINGLES/COUPLES	Premium	43390	1	1
1	1002	YOUNG SINGLES/COUPLES	Mainstream	43359	1	2
2	1003	YOUNG FAMILIES	Budget	43531	1	3
3	1003	YOUNG FAMILIES	Budget	43532	1	4
4	1004	OLDER SINGLES/COUPLES	Mainstream	43406	1	5
5	1005	MIDAGE SINGLES/COUPLES	Mainstream	43462	1	6
6	1007	YOUNG SINGLES/COUPLES	Budget	43438	1	7
7	1007	YOUNG SINGLES/COUPLES	Budget	43439	1	8
8	1009	NEW FAMILIES	Premium	43424	1	9
9	1010	YOUNG SINGLES/COUPLES	Mainstream	43352	1	10

In [32]:

```
counts = merged_data.PROD_NAME.value_counts().head(20)


colors = px.colors.qualitative.Set1[:len(counts)]

# Create a bar chart using Plotly Express with different colors
fig = px.bar(
    y=counts.index,
    x=counts.values,
    orientation='h',
    labels={'x': 'Count', 'y':'Product Name'},
    color=counts.index,
    color_discrete_map={ctype: color for ctype, color in zip(counts.index, colors)}
    title='Top - 20 Product Names (Ascending Order)',
)
for i, count in enumerate(counts.values):
    fig.add_annotation(
        y=counts.index[i],
```

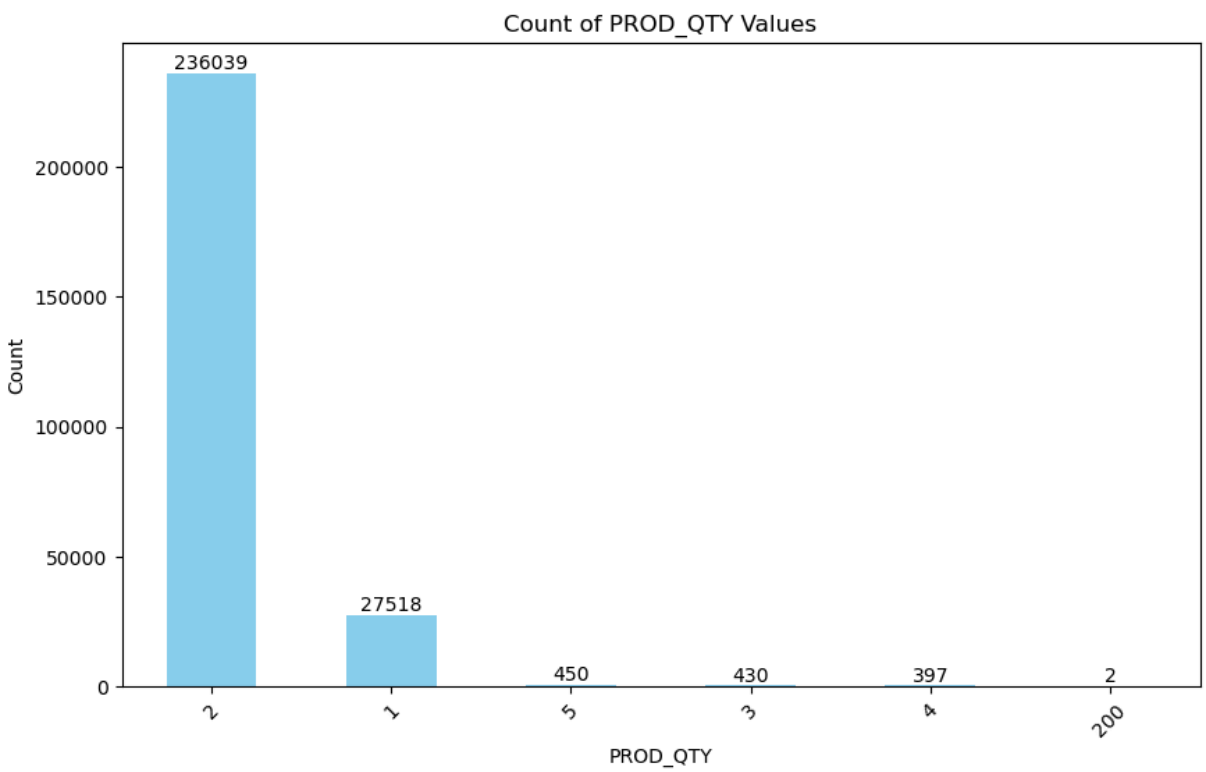
```
        x=count,
        text=str(count),
        showarrow=False,
        arrowhead=5,
        ax=0,
        ay=-30,
    )

# Add a legend
fig.update_layout(legend=dict(title=dict(text='Legend')), height = 1000)

# Show the plot
fig.show()
```


Top - 20 Product Names (Ascending Order)
Tyrrells Crisps Lightly Salted 165g
Pringles Sweet&Spcy BBQ 134g
Dorito Corn Chp Supreme 380g
Thins Chips Light& Tangy 175g
Smiths Crinkle Chips Salt & Vinegar 330g
Kettle Sweet Chilli And Sour Cream 175g
Doritos Corn Chip Mexican Jalapeno 150g
Pringles Barbeque 134g

In [33]:
```
merged_data.PROD_QTY.value_counts()
```

Out[33]:
```
2      236039
1       27518
5         450
3         430
4         397
200         2
Name: PROD_QTY, dtype: int64
```

In [34]:
```
import matplotlib.pyplot as plt
prod_qty_counts = merged_data['PROD_QTY'].value_counts()

# Plotting the counts
plt.figure(figsize=(10, 6))
ax = prod_qty_counts.plot(kind='bar', color='skyblue')
plt.title('Count of PROD_QTY Values')
plt.xlabel('PROD_QTY')
plt.ylabel('Count')
plt.xticks(rotation=45)
for i, count in enumerate(prod_qty_counts):
    ax.text(i, count + 0.1, str(count), ha='center', va='bottom')


# Display the plot
plt.show()
```



In [35]:
```
merged_data.head(5)
```

Out[35]:

	LYLTY_CARD_NBR	LIFESTAGE	PREMIUM_CUSTOMER	DATE	STORE_NBR	TXN_ID
0	1000	YOUNG SINGLES/COUPLES	Premium	43390	1	1
1	1002	YOUNG SINGLES/COUPLES	Mainstream	43359	1	2
2	1003	YOUNG FAMILIES	Budget	43531	1	3
3	1003	YOUNG FAMILIES	Budget	43532	1	4
4	1004	OLDER SINGLES/COUPLES	Mainstream	43406	1	5

In [36]:

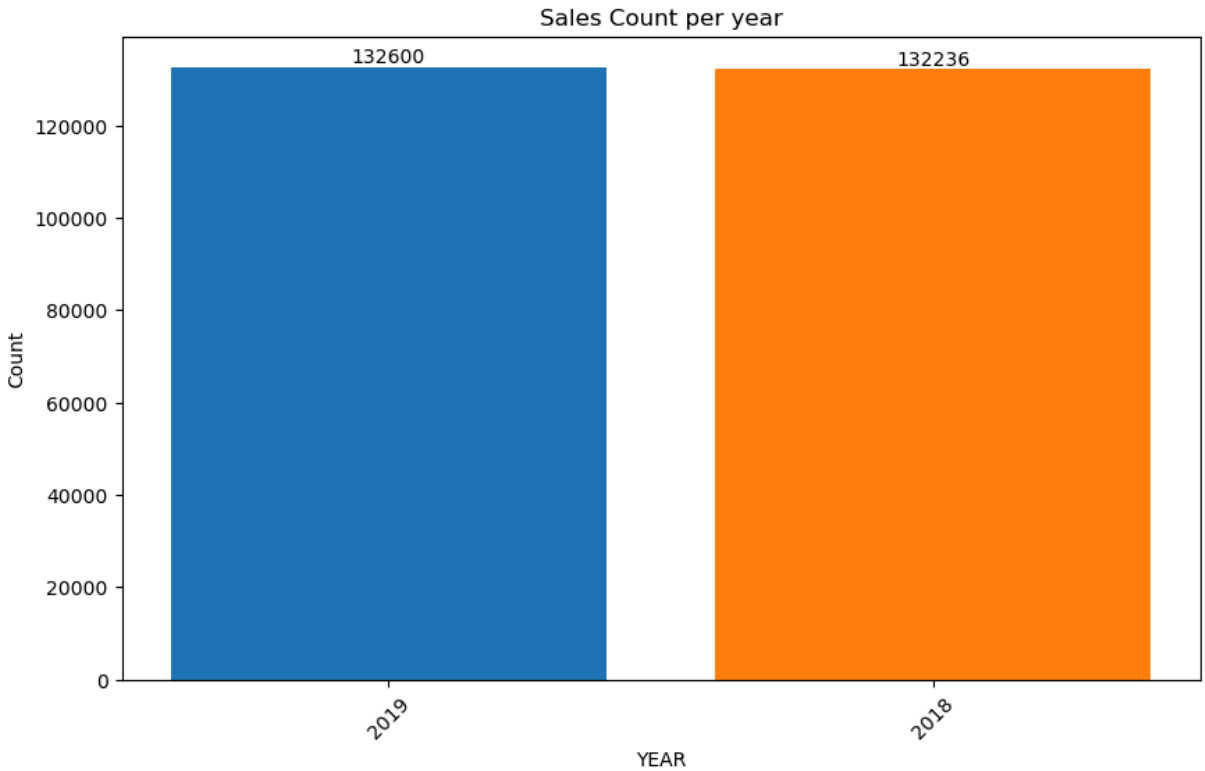
```
merged_data.Year.value_counts()
```

Out[36]:

```
2019    132600
2018    132236
Name: Year, dtype: int64
```

In [37]:

```
counts = merged_data['Year'].value_counts()

# Plotting the counts
plt.figure(figsize=(10, 6))

for i, (value, count) in enumerate(counts.items()):
    plt.bar(i, count)
    plt.text(i, count + 0.1, str(count), ha='center', va='bottom')

plt.title('Sales Count per year')
plt.xlabel('YEAR')
plt.ylabel('Count')
plt.xticks(range(len(counts)), counts.index, rotation=45)

# Display the plot
plt.show()
```

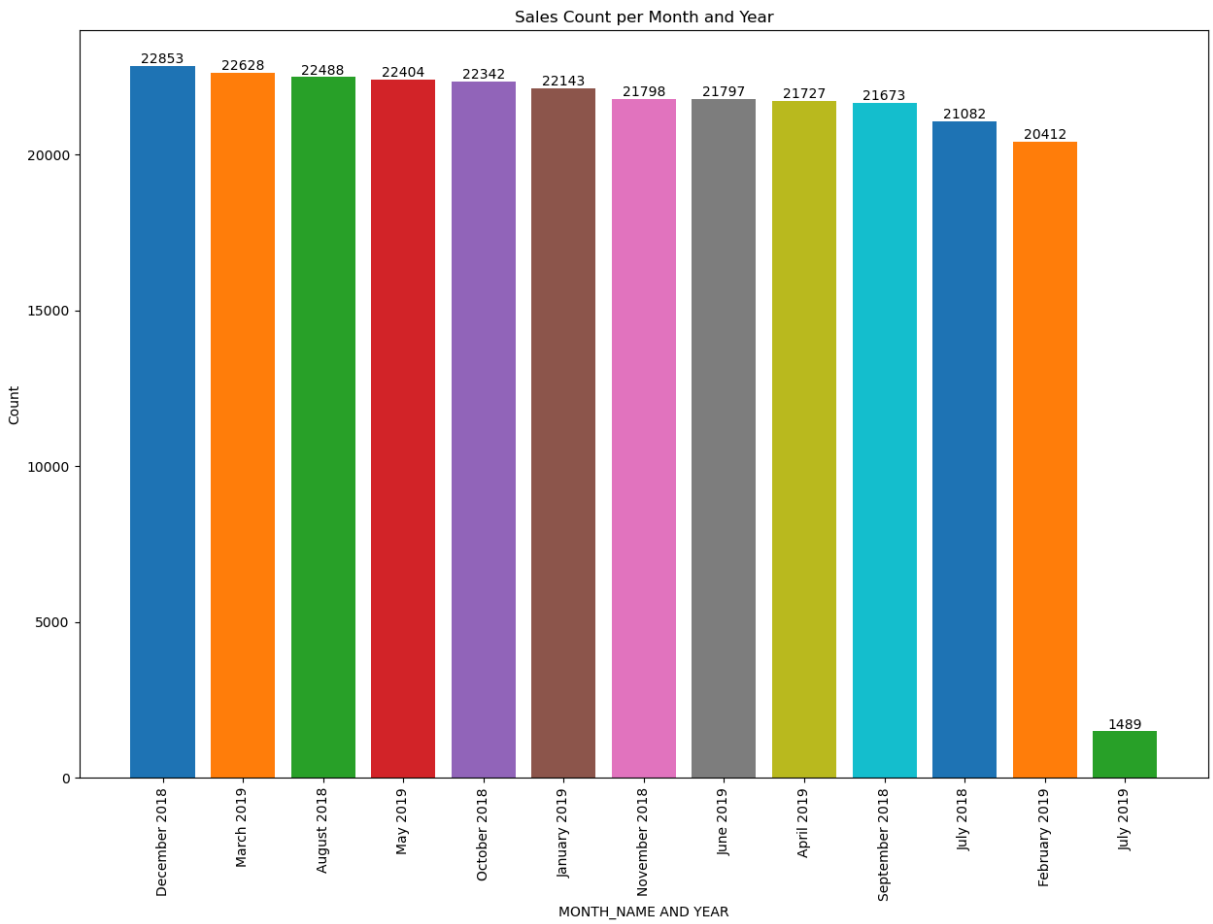


In [38]:
```
merged_data.Month_Year.value_counts()
```

Out[38]:
```
December 2018     22853
March 2019        22628
August 2018       22488
May 2019          22404
October 2018      22342
January 2019      22143
November 2018     21798
June 2019         21797
April 2019        21727
September 2018    21673
July 2018         21082
February 2019     20412
July 2019          1489
Name: Month_Year, dtype: int64
```

In [39]:
```
counts = merged_data['Month_Year'].value_counts()

# Plotting the counts
plt.figure(figsize=(15, 10))

for i, (value, count) in enumerate(counts.items()):
    plt.bar(i, count)
    plt.text(i, count + 0.1, str(count), ha='center', va='bottom')

plt.title('Sales Count per Month and Year')
plt.xlabel('MONTH_NAME AND YEAR')
plt.ylabel('Count')
plt.xticks(range(len(counts)), counts.index, rotation=90)
```

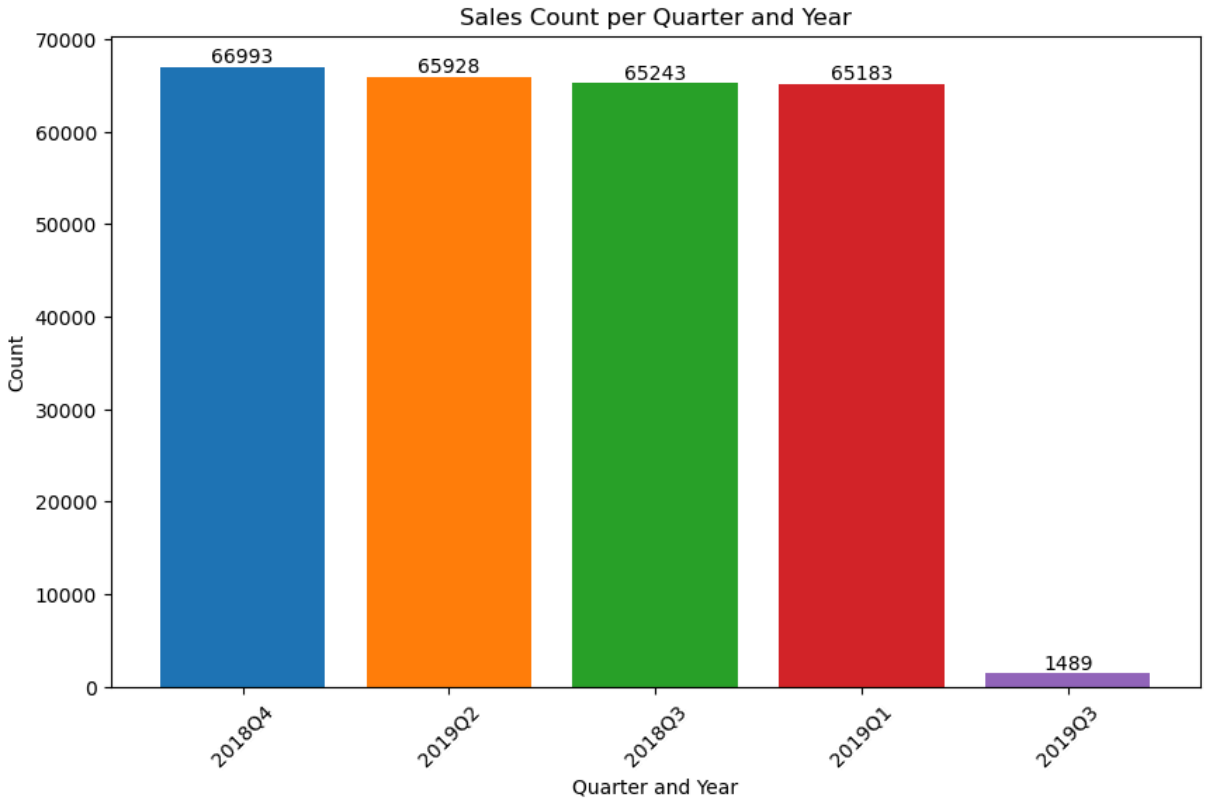
```
# Display the plot
plt.show()
```



In [40]:
```
merged_data['Quarter_Year'].value_counts()
```

Out[40]:
```
2018Q4    66993
2019Q2    65928
2018Q3    65243
2019Q1    65183
2019Q3     1489
Freq: Q-DEC, Name: Quarter_Year, dtype: int64
```

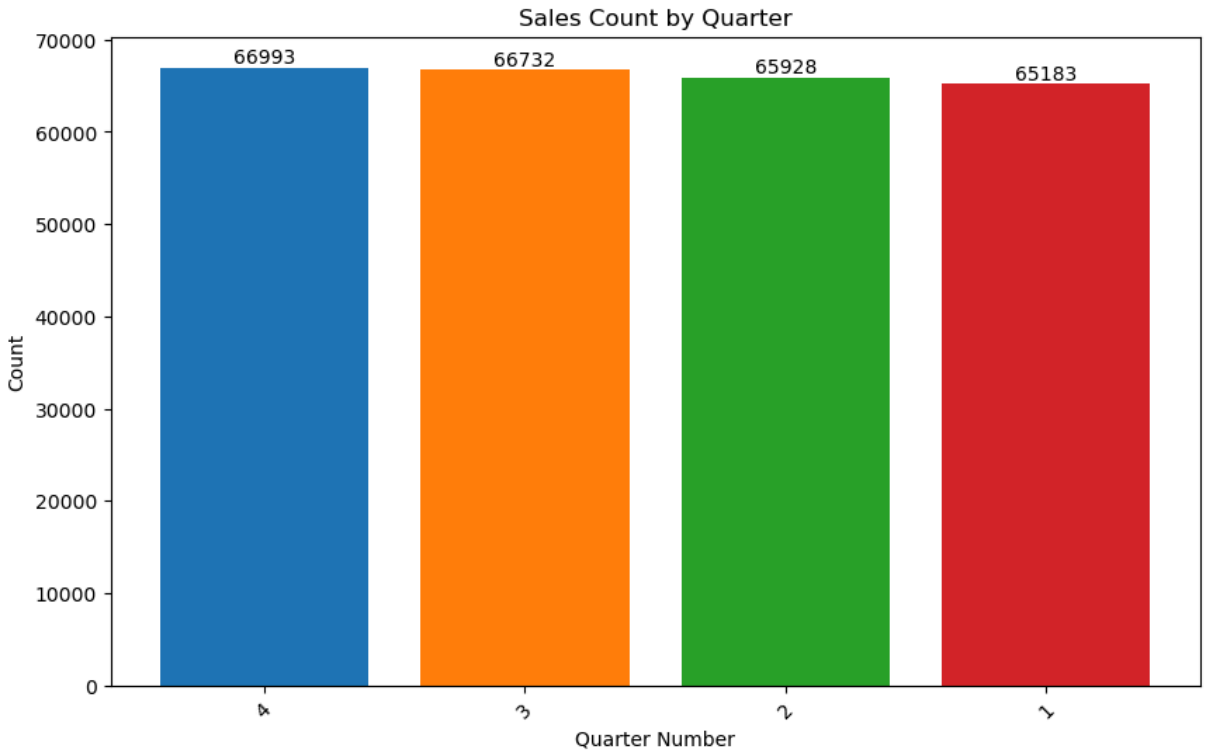
In [41]:
```
counts = merged_data['Quarter_Year'].value_counts()

# Plotting the counts
plt.figure(figsize=(10, 6))

for i, (value, count) in enumerate(counts.items()):
    plt.bar(i, count)
    plt.text(i, count + 0.1, str(count), ha='center', va='bottom')

plt.title('Sales Count per Quarter and Year')
plt.xlabel('Quarter and Year')
plt.ylabel('Count')
plt.xticks(range(len(counts)), counts.index, rotation=45)

# Display the plot
plt.show()
```



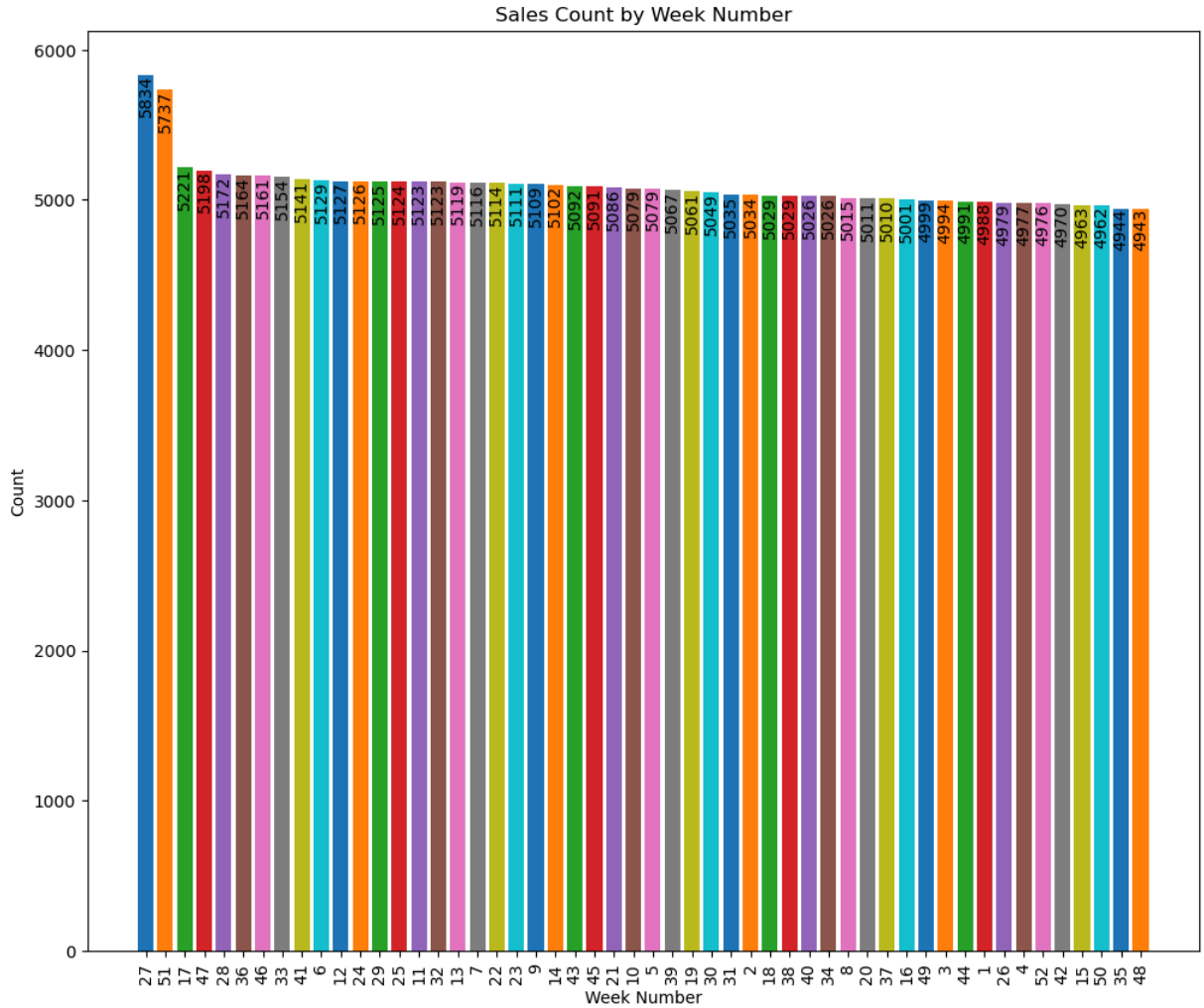
In [42]:
```
merged_data['Quarter'].value_counts()
```

Out[42]:
```
4    66993
3    66732
2    65928
1    65183
Name: Quarter, dtype: int64
```

In [43]:
```
counts = merged_data['Quarter'].value_counts()

# Plotting the counts
plt.figure(figsize=(10, 6))

for i, (value, count) in enumerate(counts.items()):
    plt.bar(i, count)
    plt.text(i, count + 0.1, str(count), ha='center', va='bottom')

plt.title('Sales Count by Quarter')
plt.xlabel('Quarter Number')
plt.ylabel('Count')
plt.xticks(range(len(counts)), counts.index, rotation=45)

# Display the plot
plt.show()
```



In [44]:
```
counts = merged_data['Week_Number'].value_counts()

# Plotting the counts
plt.figure(figsize=(12, 10))

for i, (value, count) in enumerate(counts.items()):
    plt.bar(i, count)
    plt.text(i, count + 0.1, str(count), ha='center', va='top', rotation = 90)

plt.title('Sales Count by Week Number')
plt.xlabel('Week Number')
plt.ylabel('Count')
plt.xticks(range(len(counts)), counts.index, rotation=90)

# Display the plot
plt.show()
```

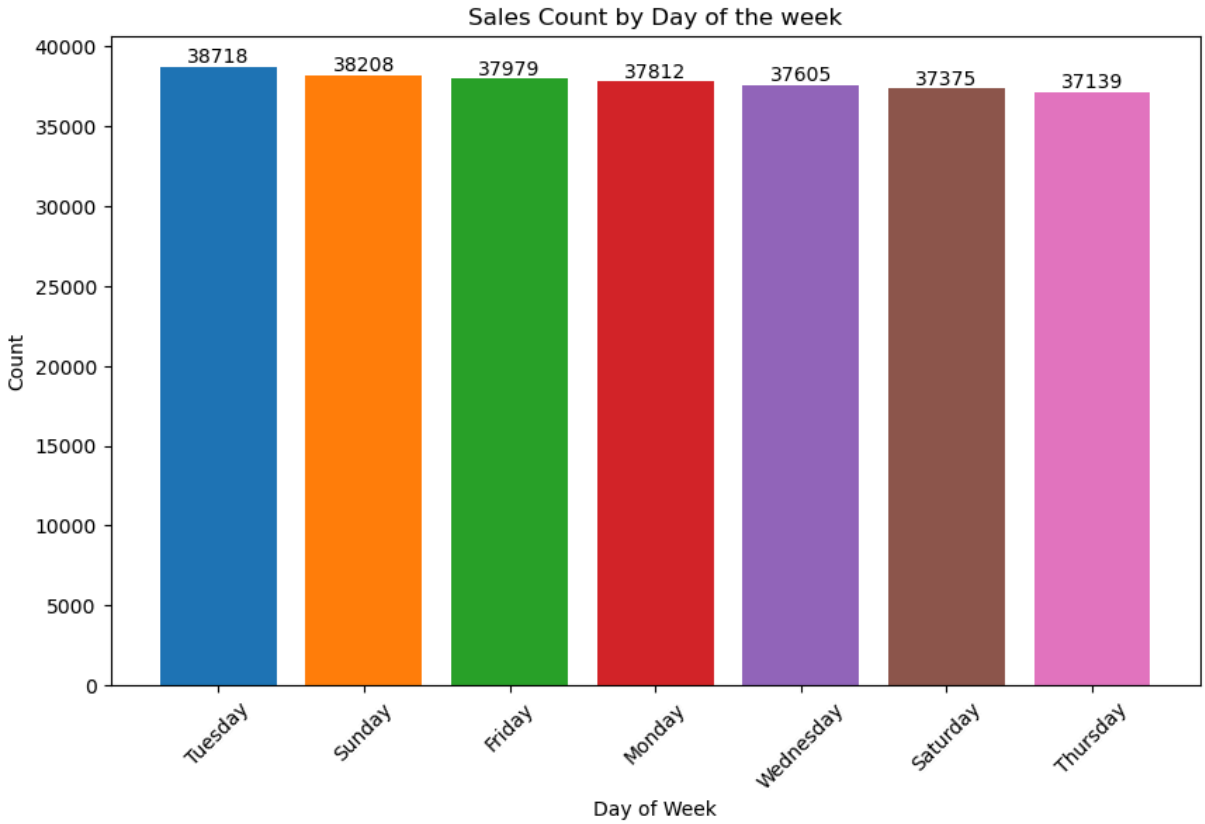


In [45]:
```
merged_data['Weekday'].value_counts()
```

Out[45]:
```
Tuesday      38718
Sunday       38208
Friday       37979
Monday       37812
Wednesday    37605
Saturday     37375
Thursday     37139
Name: Weekday, dtype: int64
```

In [46]:
```
counts = merged_data['Weekday'].value_counts()

# Plotting the counts
plt.figure(figsize=(10, 6))

for i, (value, count) in enumerate(counts.items()):
    plt.bar(i, count)
    plt.text(i, count + 0.1, str(count), ha='center', va='bottom')

plt.title('Sales Count by Day of the week')
plt.xlabel('Day of Week')
plt.ylabel('Count')
plt.xticks(range(len(counts)), counts.index, rotation=45)
```

```
# Display the plot
plt.show()
```



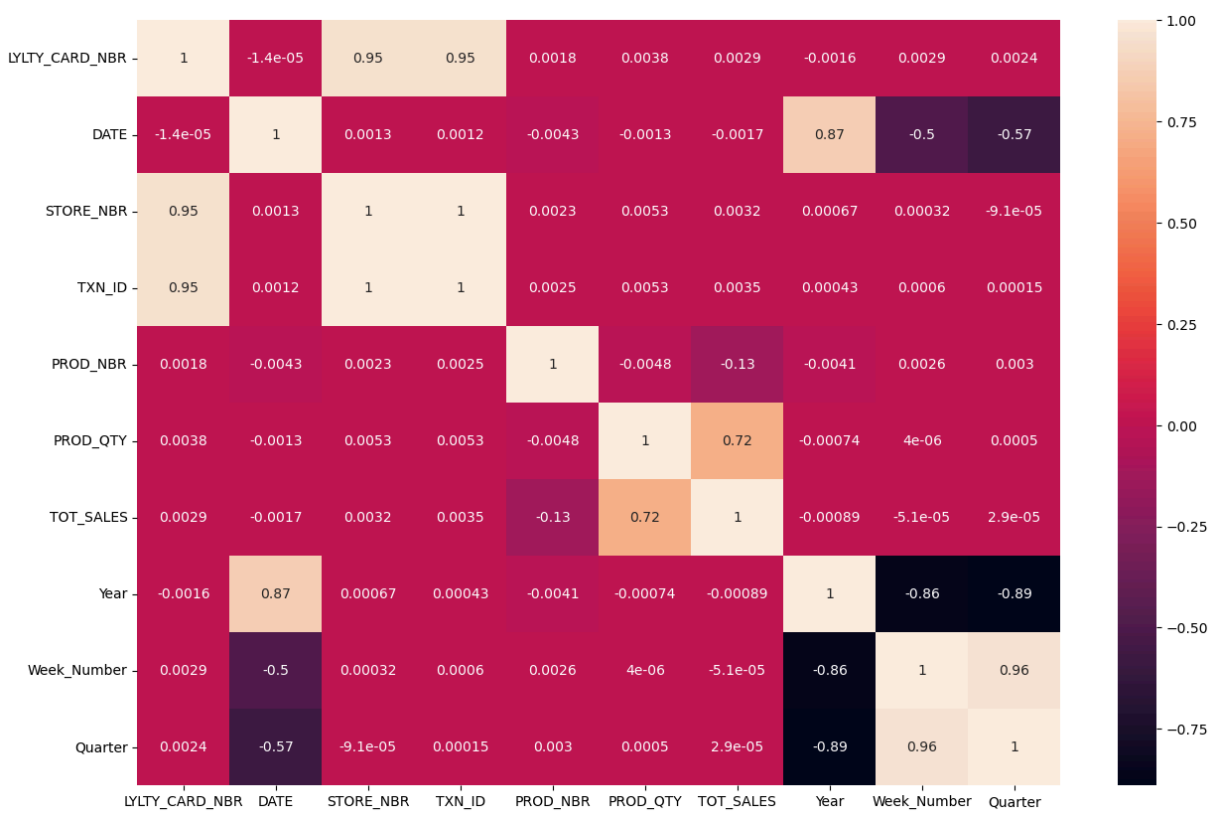
In [47]:
```
merged_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 264836 entries, 0 to 264835
Data columns (total 18 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   LYLTY_CARD_NBR    264836 non-null  int64
 1   LIFESTAGE         264836 non-null  object
 2   PREMIUM_CUSTOMER  264836 non-null  object
 3   DATE              264836 non-null  int64
 4   STORE_NBR         264836 non-null  int64
 5   TXN_ID            264836 non-null  int64
 6   PROD_NBR          264836 non-null  int64
 7   PROD_NAME         264836 non-null  object
 8   PROD_QTY          264836 non-null  int64
 9   TOT_SALES         264836 non-null  float64
 10  DATE_NRML         264836 non-null  datetime64[ns]
 11  Year              264836 non-null  int64
 12  Month_Name        264836 non-null  object
 13  Month_Year        264836 non-null  object
 14  Week_Number       264836 non-null  UInt32
 15  Quarter           264836 non-null  int64
 16  Quarter_Year      264836 non-null  period[Q-DEC]
 17  Weekday           264836 non-null  object
dtypes: UInt32(1), datetime64[ns](1), float64(1), int64(8), object(6), period[Q-DEC]
(1)
memory usage: 37.6+ MB
```

In [48]:

```
plt.figure(figsize=(15, 10))
sns.heatmap(merged_data.corr(), annot=True)
```

Out[48]: <AxesSubplot:>



In [49]:
```
merged_data.head()
```

Out[49]:

	LYLTY_CARD_NBR	LIFESTAGE	PREMIUM_CUSTOMER	DATE	STORE_NBR	TXN_ID
0	1000	YOUNG SINGLES/COUPLES	Premium	43390	1	1
1	1002	YOUNG SINGLES/COUPLES	Mainstream	43359	1	2
2	1003	YOUNG FAMILIES	Budget	43531	1	3
3	1003	YOUNG FAMILIES	Budget	43532	1	4
4	1004	OLDER SINGLES/COUPLES	Mainstream	43406	1	5